

$$\begin{pmatrix} \frac{1}{2} & 3 \\ 2 & -1 \\ 7 & 9 \end{pmatrix}_{3 \times 2}$$

$$\begin{pmatrix} 2 & -10 \\ 6 & 14 \end{pmatrix}_{2 \times 2}$$

$$\begin{pmatrix} 9 & -5 \\ 1 & 7 \end{pmatrix}_{2 \times 2}$$

$$\begin{pmatrix} 14 & 21 \\ 30 & 47 \\ 46 & 73 \end{pmatrix}_{3 \times 2}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2 \times 2}$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$3x_1 - x_2 + x_3 = 6$$

$$-x_1 + 3x_2 + 4x_3 = 4$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ 0 & 2 & 2 & 2 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ 0 & 0 & -5 & -10 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 0 & 2 & 7 & 12 \\ 0 & 0 & -5 & -10 \end{array} \right)_{3 \times 4}$$

$$x_1 = 1$$

$$x_2 = -1$$

$$x_3 = 2$$

$$3x_1 + 2x_2 + 2x_3 - 5x_4 = 8$$

$$6x_1 + 15x_2 + 15x_3 - 54x_4 = 27$$

$$12x_1 - 3x_2 - 3x_3 + 24x_4 = 21$$

Gauss Elimination:

Echelon Form:

$$\left(\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 6 & 15 & 15 & -54 & 27 \\ 12 & -3 & -3 & 24 & 21 \end{array} \right)_{3 \times 5}$$

$$\left(\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0 & 11 & 11 & -44 & 11 \\ 0 & -11 & -11 & 44 & -11 \end{array} \right)_{3 \times 5}$$

$$\left(\begin{array}{cccc|c} 3 & 2 & 2 & -5 & 8 \\ 0 & 11 & 11 & -44 & 11 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)_{3 \times 5}$$

Infinitely many solutions

$$3x_1 + 2x_2 + x_3 = 3$$

$$2x_1 + x_2 + x_3 = 0$$

$$6x_1 + 2x_2 + 4x_3 = 6$$

Gauss Elimination:

Echelon Form:

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & \frac{1}{3} & -2 \\ 0 & -2 & 2 & 0 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & \frac{1}{3} & -2 \\ 0 & 0 & 0 & 12 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & \frac{1}{3} & -2 \\ 0 & 0 & 0 & 12 \end{array} \right)_{3 \times 4}$$

No solution

$$i_1 - i_2 + i_3 = 0$$

$$2i_2 + i_3 = 8$$

$$2i_2 + 5i_3 = 18$$

Gauss Elimination:

Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 8 \\ 0 & 2 & 5 & 18 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 8 \\ 0 & 2 & 5 & 18 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 8 \\ 0 & 0 & 4 & 10 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 8 \\ 0 & 0 & 4 & 10 \end{array} \right)_{3 \times 4}$$

$$i_1 = \frac{1}{4}$$

$$i_2 = \frac{11}{4}$$

$$i_3 = \frac{5}{2}$$

Echelon Form:

$$\left(\begin{array}{cccc} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{cccc} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{array} \right)_{3 \times 4}$$

$$\left(\begin{array}{cccc} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{array} \right)_{3 \times 4}$$

$$3x + 7y + 8z = -13$$

$$2x + 9z = -5$$

$$-4x + y - 26z = 2$$

Determinant:

Echelon Form:

$$\begin{vmatrix} 3 & 7 & 8 \\ 2 & 0 & 9 \\ -4 & 1 & -26 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & 8 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & \frac{31}{3} & -\frac{46}{3} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & 8 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & 0 & -\frac{101}{14} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & 8 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & 0 & -\frac{101}{14} \end{vmatrix}_{3 \times 3}$$

Determinant:

Echelon Form:

$$\begin{vmatrix} -13 & 7 & 8 \\ -5 & 0 & 9 \\ 2 & 1 & -26 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} -13 & 7 & 8 \\ 0 & -\frac{35}{13} & \frac{77}{13} \\ 0 & \frac{27}{13} & -\frac{322}{13} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} -13 & 7 & 8 \\ 0 & -\frac{35}{13} & \frac{77}{13} \\ 0 & 0 & -\frac{101}{5} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} -13 & 7 & 8 \\ 0 & -\frac{35}{13} & \frac{77}{13} \\ 0 & 0 & -\frac{101}{5} \end{vmatrix}_{3 \times 3}$$

Determinant:

Echelon Form:

$$\begin{vmatrix} 3 & -13 & 8 \\ 2 & -5 & 9 \\ -4 & 2 & -26 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & -13 & 8 \\ 0 & \frac{11}{3} & \frac{11}{3} \\ 0 & -\frac{46}{3} & -\frac{46}{3} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & -13 & 8 \\ 0 & \frac{11}{3} & \frac{11}{3} \\ 0 & 0 & 0 \end{vmatrix}_{3 \times 3}$$

Determinant:
Echelon Form:

$$\begin{vmatrix} 3 & 7 & -13 \\ 2 & 0 & -5 \\ -4 & 1 & 2 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & -13 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & \frac{31}{3} & -\frac{46}{3} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & -13 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & 0 & -\frac{101}{14} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 3 & 7 & -13 \\ 0 & -\frac{14}{3} & \frac{11}{3} \\ 0 & 0 & -\frac{101}{14} \end{vmatrix}_{3 \times 3}$$

$$x = -7$$

$$y = 0$$

$$z = 1$$

Determinant:
Echelon Form:

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 7 \\ 0 & 2 & 2 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 7 \\ 0 & 0 & -5 \end{vmatrix}_{3 \times 3}$$

$$\begin{pmatrix} -1 & 1 & 2 \\ 0 & 2 & 7 \\ 0 & 0 & -5 \end{pmatrix}_{3 \times 3}$$

Inverse:

$$\begin{pmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{pmatrix}_{3 \times 3}$$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right)_{3 \times 6}$$

Echelon Form:

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 0 & -5 & -4 & -1 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 0 & -5 & -4 & -1 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 0 & -\frac{3}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 2 & 0 & -\frac{13}{5} & -\frac{2}{5} & \frac{7}{5} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} -1 & 0 & 0 & \frac{7}{10} & -\frac{1}{5} & -\frac{3}{10} \\ 0 & 1 & 0 & -\frac{13}{10} & -\frac{1}{5} & \frac{7}{10} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{7}{10} & \frac{1}{5} & \frac{3}{10} \\ 0 & 1 & 0 & -\frac{13}{10} & -\frac{1}{5} & \frac{7}{10} \\ 0 & 0 & 1 & \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{array} \right)_{3 \times 6}$$

$$\begin{pmatrix} -\frac{7}{10} & \frac{1}{5} & \frac{3}{10} \\ -\frac{13}{10} & -\frac{1}{5} & \frac{7}{10} \\ \frac{4}{5} & \frac{1}{5} & -\frac{1}{5} \end{pmatrix}_{3 \times 3}$$

Determinant:
Echelon Form:

$$\begin{vmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 0 & 1 & 3 \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} \end{vmatrix}_{3 \times 3}$$

$$\begin{vmatrix} 2 & 0 & -1 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} \end{vmatrix}_{3 \times 3}$$

Inverse:

$$\begin{pmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix}_{3 \times 3}$$

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right)_{3 \times 6}$$

Echelon Form:

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{5}{2} & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{5}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{5}{2} & -1 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{5}{2} & 1 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{5}{2} & -1 & 1 \end{array} \right)_{3 \times 6}$$

$$\left(\begin{array}{ccc|ccc}2&0&0&6&-2&2\\0&1&0&-15&6&-5\\0&0&1&5&-2&2\end{array}\right)_{3\times 6}$$

$$\left(\begin{array}{ccc|ccc}2&0&0&6&-2&2\\0&1&0&-15&6&-5\\0&0&1&5&-2&2\end{array}\right)_{3\times 6}$$

$$\left(\begin{array}{ccc|ccc}1&0&0&3&-1&1\\0&1&0&-15&6&-5\\0&0&1&5&-2&2\end{array}\right)_{3\times 6}$$

$$\left(\begin{array}{ccc}3&-1&1\\-15&6&-5\\5&-2&2\end{array}\right)_{3\times 3}$$

$$\nabla\left(x^2+3y^2-5z^2\right)=\left(\begin{array}{c}2x\\6y\\-10z\end{array}\right)$$

$$\left(\begin{array}{c}2x\\6y\\-10z\end{array}\right)$$

$$\nabla\cdot\left(\begin{array}{c}xy\\y^2+z^2\\-5xyz\end{array}\right)=-5xy+3y$$

$$-5xy+3y$$

$$\nabla\times\left(\begin{array}{c}xyz\\2xy\\x^2+y^2\end{array}\right)=\left(\begin{array}{c}2y\\-2x+xy\\-xz+2y\end{array}\right)$$

$$\left(\begin{array}{c}2y\\-2x+xy\\-xz+2y\end{array}\right)$$

$$\frac{d}{dx}\left(\begin{array}{c}1\\2\\3\end{array}\right)=\left(\begin{array}{c}0\\0\\0\end{array}\right)$$

$$\left(\begin{array}{c}0\\0\\0\end{array}\right)$$

$$\frac{d}{d \begin{pmatrix} x \\ y \\ z \end{pmatrix}} 5 = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

$$\frac{d}{d \begin{pmatrix} x & y & z \\ x_1 & x_2 & x_3 \\ i_1 & i_2 & i_3 \end{pmatrix}_{3 \times 3}} 5 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{3 \times 3}$$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{3 \times 3}$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{3 \times 9}$$

$$\frac{d}{d \begin{pmatrix} x \\ y \\ z \\ x_1 \\ x_2 \\ x_3 \\ i_1 \\ i_2 \\ i_3 \end{pmatrix}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}_{9 \times 9}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}_{9 \times 9}$$

$$\frac{d}{d\begin{pmatrix} x \\ y \\ z \end{pmatrix}} x + 2y + 3z = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$\frac{d}{d\begin{pmatrix} x \\ y \\ z \end{pmatrix}} \begin{pmatrix} x + 2y + 3z \\ 3x + 4y + 6z \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \end{pmatrix}_{2 \times 3}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \end{pmatrix}_{2 \times 3}$$